

Measuring water quality with biodiversity

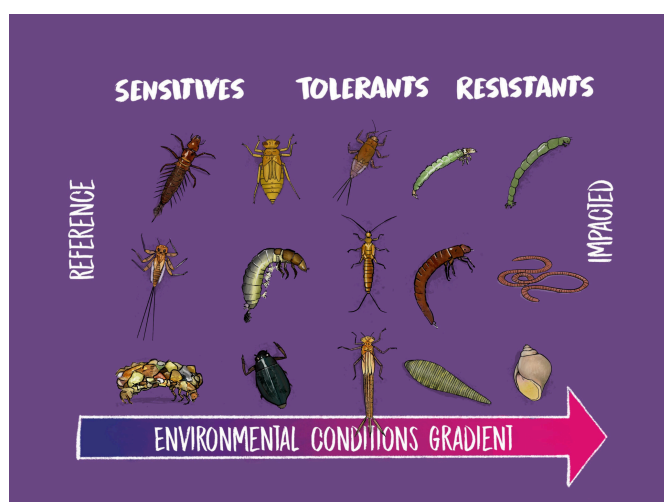
This dataset and the following information was provided by Data Classroom, Inc., and is available for free online for educational use.

Background

There are many ways to measure environmental quality and the health of an ecosystem. The use of **biological assessment** is often both a practical and powerful way to measure the health of an ecosystem. One common biological assessment used to measure the health of streams and small rivers around the world is sampling benthic macroinvertebrates. These *bottom* (benthos) dwelling animals are large enough to be seen *without a microscope* (macro) and *do not have a backbone* (invertebrate). Macroinvertebrate sampling can reveal the physical and or chemical health of stream because the presence of specific animals known to be intolerant to disturbance and pollution can give an overall assessment of ecosystem health.

Benthic macroinvertebrates can be easily sampled in a small stream by wading in shallow water and dragging a handheld net along the bottom. Commonly found animal groups collected with this method include crustaceans (crayfish, shrimp), insects (mayflies, caddisflies, stoneflies, midges, beetles) , worms, snails, and clams. The presence of these small animals is important because their presence indicates that the waterbody where they are found, as part of the larger watershed, is able to support complex food webs from primary producers to larger consumers such as fish.

Mayflies (**E**phemeroptera), stoneflies (**P**lecoptera), and caddisflies (**T**richoptera) are known to be particularly sensitive to pollution and disturbance, and their presence in a stream indicates an ecosystem that can support a diverse community of organisms. The



number of such insects is often used to generate a score called the EPT index. In general, a high level of species **richness** (a measure of how many different species live together in one place) indicates an ecosystem that can support ample biodiversity to indicate overall watershed health. If one particular animal group is a highly **dominant taxa** because it makes up a high percentage of all the animals found there, it often indicates that the ecosystem is disturbed or polluted.

Since 1999 the Pennsylvania Department of Environmental Protection (PDEP) has been using the same methods for benthic macroinvertebrate sampling to measure the health of their streams across the state. In addition to sampling the macroinvertebrates, the PDEP and their partners have been collecting data on habitat quality and characteristics of streams like the amount of vegetation along the stream bank, known as a **riparian buffer**, and the fast-moving shallow sections of the stream called **riffles** which increase the oxygen in the water. They use habitat use data like those to compile an overall **Total Habitat Score** where higher numbers indicate a less disturbed waterway.

Dataset

This dataset is just a small part of all the data that was collected by PDEP. It is made up of data collection from a single year (2016) and was collected only from streams known as freestone with watersheds of similar size (between 1 and 25 square miles). Additionally, we eliminated any samples with a total habitat score of 0.

Variables

StationID - which individual field stations in Pennsylvania recorded the data for an individual observation (row) in the dataset.

Richness - number of different species found in the sample.

Shannon Diversity Index - This index was developed to quantify the richness and evenness of a biological community of organisms. Typical values are generally between 1.5 and 3.5 in most ecological studies, and the index is rarely greater than 4. Higher values indicate greater biodiversity.

% Intolerant Mayfly - percentage of the sample that was made up of individuals that were mayflies. Mayfly species are known to be intolerant of pollution.

% Dominant Taxa - percentage of the total individuals in the sample that were from the most frequently found species in the sample.